3–15 (a) Formulate mesh-current equations for the circuit in Figure P3–15.
(b) Formulate node-voltage equations for the circuit in Figure P3–15.
(c) Which set of equations would be easier to solve? Why?
(d) Find v_x and i_x using whichever method you prefer.

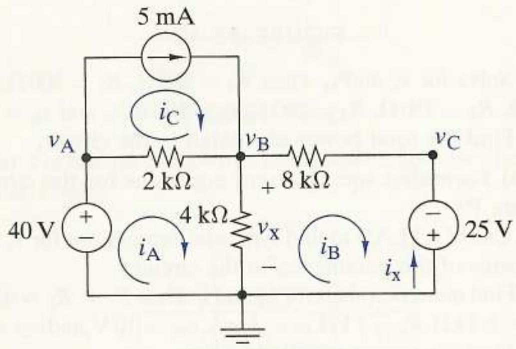


FIGURE P3–15

3–16 Use mesh-current equations, node-voltage equations, or simple engineering intuition to find the input resistance of the circuit in Figure P3–16.

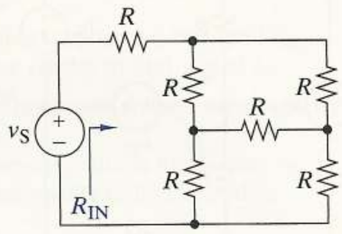


FIGURE P3–16

3–17 In Figure P3–17 all of the resistors are 1 kΩ and $v_S = 10$ V. The voltage at node C is found to be $v_C = -2$ V when node B is connected to ground. Find the node voltages v_A and v_D, and the mesh currents i_A and i_B.

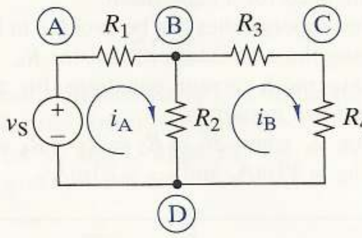


FIGURE P3–17

3–18 Use Figure P3–17 and MATLAB to solve the following problems:
(a) Using mesh-current analysis, find a symbolic expression for i_A in terms of the circuit parameters.
(b) Compute the ratio v_S/i_A.
(c) Find a symbolic expression for the equivalent resistanc of the circuit by combining resistors in series and paralle Compare your answer to the results from part (b).

3–19 Use OrCAD to find the node voltages v_A and v_B and th mesh currents i_A, i_B, and i_C in Figure P3–19.

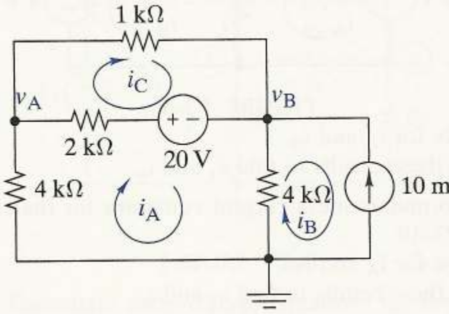


FIGURE P3–19

3–20 (a) Use MATLAB and mesh-current analysis to solve f the mesh currents i_A, i_B, i_C, and i_D in Figure P3–20.
(b) Solve for the same currents using node-voltage analy and compare the effort required with each technique.
(c) Use OrCAD to verify your results in the previous tw parts.

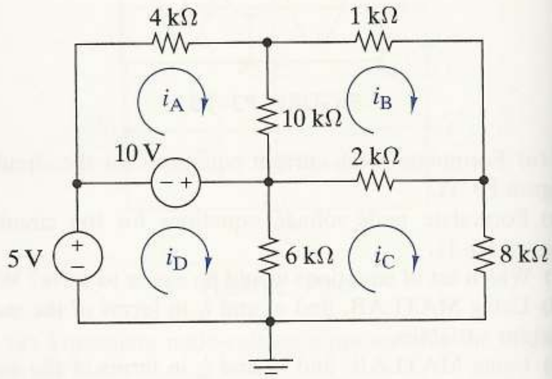


FIGURE P3–20

OBJECTIVE 3–2 LINEARITY PROPERTIES (SECT. 3–3)

(a) Given a circuit containing linear resistors and one in pendent source, use the proportionality principle to f selected signal variables.
(b) Given a circuit containing linear resistors and two or m independent sources, use the superposition principle find selected signal variables.

See Examples 3–10, 3–11, 3–12 and Exercises 3–18, 3–19, 3– 3–21, 3–22

3-21 Find the proportionality constant $K = v_O/i_S$ for the circuit in Figure P3–21.

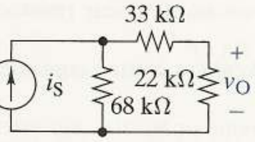


FIGURE P3–21

3-22 Find the proportionality constant $K = i_O/i_S$ for the circuit in Figure P3–22.

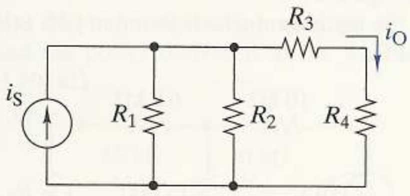


FIGURE P3–22

3-23 Find the proportionality constant $K = v_O/v_S$ for the circuit in Figure P3–23.

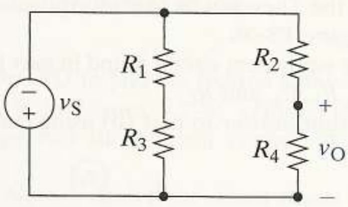


FIGURE P3–23

3-24 Use the unit output method to find v_O in Figure P3–24.

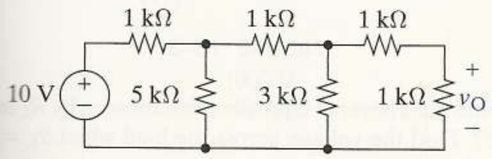


FIGURE P3–24

3-25 Use the unit output method to find i_O in Figure P3–25.

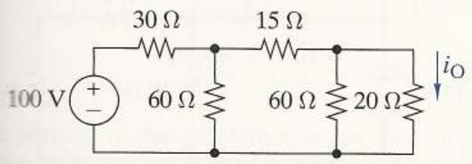


FIGURE P3–25

3-26 Use the superposition principle to find v_O in Figure P3–26.

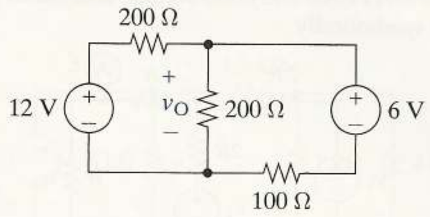


FIGURE P3–26

3-27 Use the superposition principle to find i_O in Figure P3–27. Verify your answer using OrCAD.

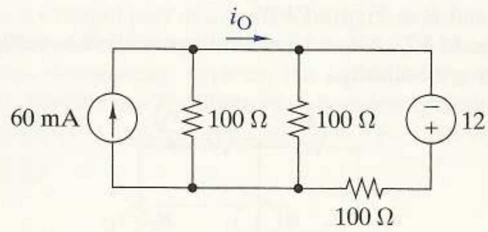


FIGURE P3–27

3-28 Use the superposition principle to find v_O in Figure P3–28.

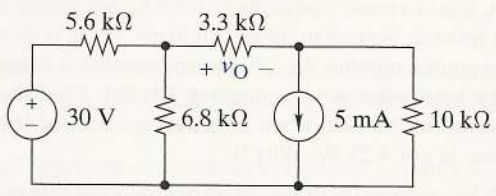


FIGURE P3–28

3-29 Use the superposition principle to find i_O in Figure P3–29. Verify your answer using OrCAD.

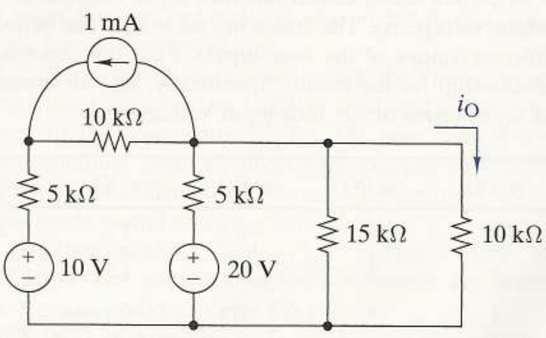


FIGURE P3–29

3–30 **(a)** Use the superposition principle to find v_O in terms of v_1, v_2, and R in Figure P3–30.
(b) Use MATLAB and node-voltage analysis to verify your answer symbolically.

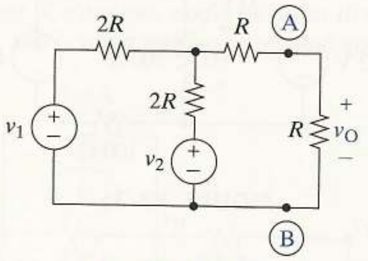


FIGURE P3–30

3–31 **(a)** Use the superposition principle to find v_O in terms of v_S, i_S, and R in Figure P3–31.
(b) Use MATLAB and node-voltage analysis to verify your answer symbolically.

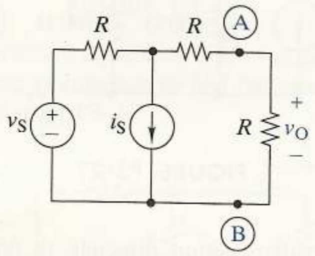


FIGURE P3–31

3–32 A linear circuit containing two sources drives a 100-Ω load resistor. Source number 1 delivers 250 mW to the load when source number 2 is off. Source number 2 delivers 4 W to the load when source number 1 is off. Find the power delivered to the load when both sources are on. (*Hint*: The answer is not 4.25 W. Why?)

3–33 A linear circuit is driven by an independent voltage source $v_S = 10$ V and an independent current source $i_S = 10$ mA. The output voltage is $v_O = 2$ V when the voltage source is on and the current source off. The output is $v_O = 1$ V when both sources are on. Find the output voltage when $v_S = 20$ V and $i_S = -20$ mA.

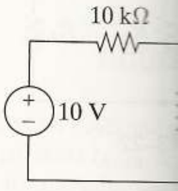
3–34 A certain linear circuit has four input voltages and one output voltage v_O. The following table lists the output for different values of the four inputs. Find the input-output relationship for the circuit. Specifically, find an expression for v_O in terms of the four input voltages.

v_{S1} (V)	v_{S2} (V)	v_{S3} (V)	v_{S4} (V)	v_O (V)
2	4	−4	1	20
1	2	2	1.5	−4
1	4	2	2	−1
0	5	3	−1	3

OBJECTIVE 3–3 THÉVE
EQUIVALENT CIRCUITS (

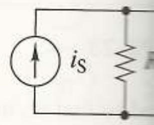
Given a circuit containing
sources,
(a) Find the Thévenin or No
of terminals.
(b) Use the Thévenin or No
delivered to linear or no
See Examples 3–13, 3–14,
3–24, 3–25, 3–26

3–35 **(a)** Find the Thévenin
by R_L in Figure P3–35.
(b) Use the equivalent cir
$R_L = 47$ kΩ.



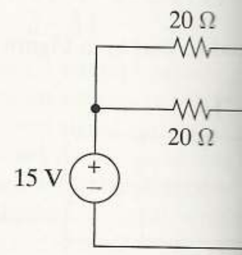
FIGU

3–36 **(a)** Find the Thévenin
by R_L in Figure P3–36.
(b) Use the equivalent circ
terms of i_S, R_1, R_2, and R_L
(c) Check your answer to



FIGUR

3–37 Find the Thévenin equiv
P3–37. Find the voltage acr
Ω, and 50 Ω.



FIGURE

SUMMARY

- A linear dependent source generates a voltage or a current whose value is proportional to a voltage or current at another point in a circuit. There are four such sources: the current-controlled voltage source, the voltage-controlled voltage source, the current-controlled current source, and the voltage-controlled current source.
- Circuits containing dependent sources can be analyzed by node-voltage or mesh-current methods. Such circuits can have input-output relationships that produce voltage, current, or power gain. The presence of feedback can dramatically influence the input and output resistances of active circuits.
- The OP AMP is an active device with five terminals called the inverting input, the noninverting input, the output, and two power supply terminals. The device is a high-gain differential amplifier with three possible operating modes: +saturation, −saturation, and linear. The output predicted by the linear mode circuit

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PROBLEMS

OBJECTIVE 4–1 LINEAR ACTIVE CIRCUITS (SECTS. 4–1, 4–2)

Given a circuit containing linear resistors, dependent sources, and independent sources, find selected output signal variables, input-output relationships, or input-output resistances.
See Examples 4–1, 4–2, 4–3, 4–4, 4–5, 4–6, 4–7, 4–8, 4–9 and Exercises 4–1, 4–2, 4–3, 4–5, 4–6, 4–7, 4–8, 4–9, 4–10, 4–11

4–1 Find the voltage gain v_O/v_S and current gain i_O/i_x in Figure P4–1 for $r = 10\,\text{k}\Omega$.

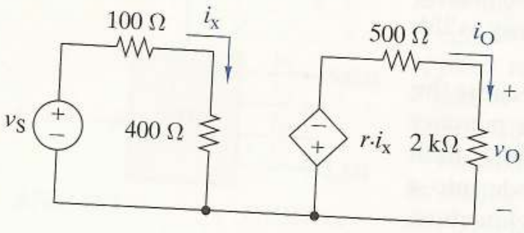
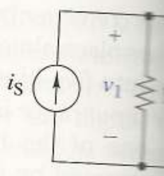
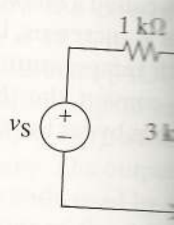


FIGURE P4–1

4–2 Find the voltage gain v_O/v_1 and the current gain i_O/i_S in Figure P4–2. For $i_S = 2\,\text{mA}$, find the power supplied by the input current source and the power delivered to the 2-kΩ load resistor.



4–3 Find the vo
Figure P4–3 f



4–4 **(a)** Find the
Figure P4–4.

Verify your answers by simulating the circuit in OrCAD.

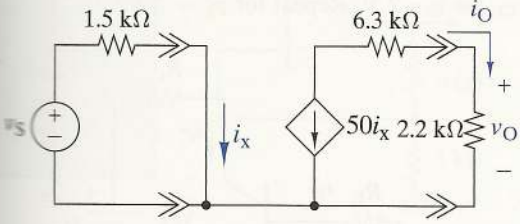


FIGURE P4–4

Find the voltage gain v_O/v_S in Figure P4–5.

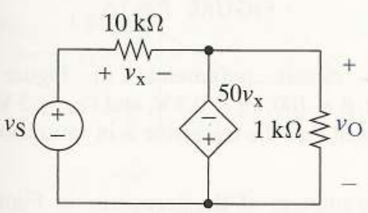


FIGURE P4–5

Find an expression for the current gain i_O/i_S in Figure P4–6. (*Hint:* Apply KCL at node A.)

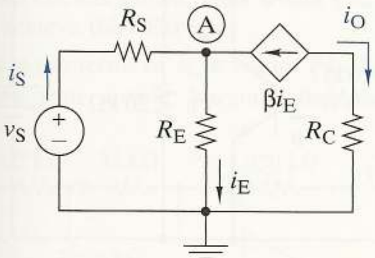


FIGURE P4–6

(a) Find the voltage v_O in Figure P4–7.

(b) Verify your answer by simulating the circuit in OrCAD.

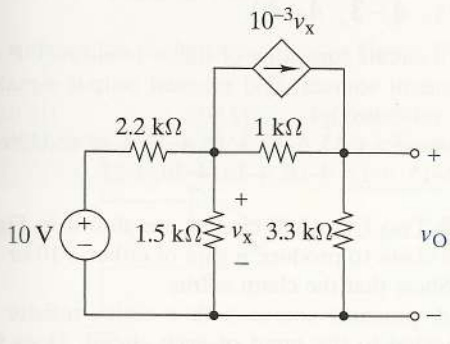


FIGURE P4–7

4–8 **(a)** Find an expression for the current gain i_O/i_S in Figure P4–8.

(b) Find an expression for the voltage gain v_O/v_S in Figure P4–8.

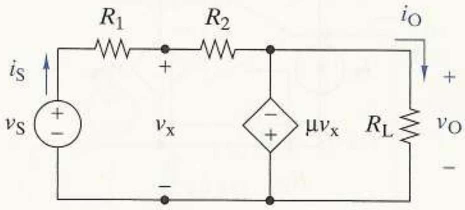


FIGURE P4–8

4–9 Find an expression for the voltage gain v_O/v_S in Figure P4–9.

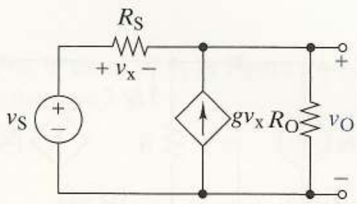


FIGURE P4–9

4–10 **(a)** Find an expression for the voltage gain v_O/v_S in Figure P4–10.

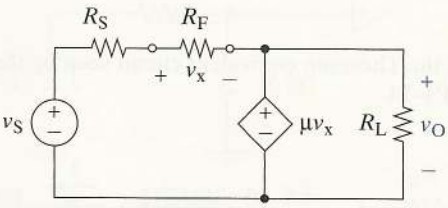


FIGURE P4–10

(b) Let $R_S = 10\,\text{k}\Omega$, $R_L = 10\,\text{k}\Omega$, and $\mu = 100$. Find the voltage gain v_O/v_S as a function of R_F. What is the voltage gain when R_F is an open circuit, when R_F is a short circuit, and for $R_F = 100\,\Omega$?

(c) Simulate the circuit in OrCAD by varying R_F from $1\,\Omega$ to $10\,\text{M}\Omega$. Read your output for $R_F = 100\,\Omega$. How does your answer compare with that for part (b)?

4–11 Find the Thévenin equivalent circuit that the load R_L sees in Figure P4–11.

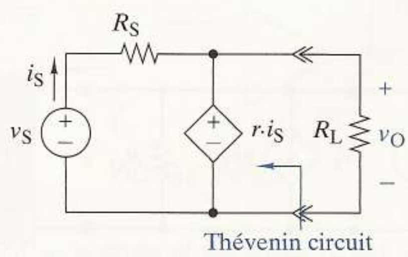


FIGURE P4–11